

Week 4: Gemini Fallback and Rescue Workflow

Gemini Fallback

The fourth week focused on handling uncertain ML predictions.

When BERTweet produces a confidence score below **0.70**, the message is forwarded to Gemini for contextual analysis.

Gemini provides additional information such as:

```
```text
```

Category

Urgency

Location

Explanation

```
This helps the system handle ambiguous messages that may not be confidently classified by BERTweet.
```

```
Low-Confidence Example
```

```
For example:
```

```
```text
```

```
"Something happened near the bridge"
```

BERTweet may produce:

```
```text
```

Category: other\_relevant\_information

Confidence: 0.54

```
Since:
```

```
```text
```

```
0.54 < 0.70
```

the system sends the message to Gemini.

Gemini then performs contextual analysis and provides additional information such as urgency, location and explanation.

Not-Humanitarian Verification

The Gemini fallback is also triggered when BERTweet predicts:

```
```text
```

not\_humanitarian

For example:

```
```text
"Beautiful weather today!"
```

Even if BERTweet produces a high confidence score, the system sends the message to Gemini for verification because it belongs to the not_humanitarian category.

This provides an additional safety check before the final classification is stored.

Rescue Team Workflow

The rescue-management functionality was also integrated.

Admins can assign incidents to rescue teams, while rescue teams can update incident status.

The system therefore connects:

```
```text
```

Reporter

↓

Incident

↓

ML Classification

↓

Admin

↓

Rescue Team

↓

Resolution

The incident model stores the assigned rescue team, status, priority, notes and status history.

### ## Security

Several security measures were also incorporated:

- JWT authentication
- Role-based access control
- Password hashing
- Input validation
- Rate limiting
- CORS protection
- Helmet security headers
- Environment variables for sensitive configuration

### ## Outcome

By the end of Week 4, the complete ML decision pipeline and rescue-management workflow were integrated